



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

AM2101-1 – MODERN COMPUTER ARCHITECTURE
(AIML/CCE)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In the case of, Zero-address instruction method the operands are stored in ____
 A) Registers B) Accumulators
 C) Push down stack D) Cache
2. The instruction, Add #45,R1 does ____
 A) Adds the value of 45 to the address of R1 and stores 45 in that address B) Adds 45 to the value of R1 and stores it in R1
 C) Finds the memory location 45 and adds that content to that of R1 D) None of the mentioned
3. ____ addressing mode is most suitable to change the normal sequence of execution of instructions.
 A) Relative B) Indirect
 C) Index with Offset D) Immediate
4. Booth algorithm gives a procedure for multiplying binary integers in ____ representation.
 A) signed-2's complement B) signed-magnitude
 C) signed-1's complement D) sign-complement
5. In the following indexed addressing mode instruction, MOV 5(R1), LOC the effective address is ____
 A) $EA = 5 + R1$ B) $EA = R1$
 C) $EA = [R1]$ D) $EA = 5 + [R1]$
6. A ripple carry adder uses ____ number of full adders for performing n-bit operation?
 A) N B) $n+1$
 C) $2n$ D) $n+2$
7. The multiplier -6(11010) is recoded as
 A) 0 -1 -2 B) 0 -1 -1
 C) -2 -1 0 D) none of the mentioned
8. In a 4-bit carry look-ahead adder the input bits are 1011 and 1001. The generate, propagate and carry signal from third stage are respectively.
 A) 0 0 0 B) 0 1 0
 C) 1 1 0 D) 1 0 1
9. Each stage in pipelining should be completed within ____ cycle.
 A) 1 B) 2
 C) 3 D) 4
10. A set of microinstructions for a single machine instruction is called ____
 A) Program B) Command
 C) Micro program D) Micro command
11. What does MAR stand for?
 A) Main Address Register B) Memory Access Register
 C) Main Accessible Register D) Memory Address Register

12. In a three BUS architecture, how many input and output ports are there in register file?
 A) 2 output and 2 input B) 1 output and 2 input
 C) 2 output and 1 input D) 1 output and 1 input
13. The main advantage of multiple bus organization over a single bus is _____.
 A) Reduction in the number of cycles for execution B) Increase in size of the registers
 C) Better Connectivity D) None of the mentioned
14. In multiple Bus organization, the registers are collectively placed and referred as _____.
 A) Set registers B) Register file
 C) Register Block D) Map registers
15. _____ are the different type/s of generating control signals.
 A) Micro-programmed B) Hardwired
 C) Micro-instruction D) Both Micro-programmed and Hardwired
16. The type of control signal is generated based on _____.
 A) contents of the step counter B) Contents of IR
 C) Contents of condition flags D) All of the mentioned
17. In super-scalar processors, _____ mode of execution is used.
 A) In-order B) Post order
 C) Out of order D) None of the mentioned
18. The method for updating the main memory as soon as a word is removed from the Cache is called.
 A) Write-through B) Write-back
 C) protected write D) Intense heat radiations
19. _____ Mapping, the data can be mapped any where in the cache memory.
 A) Associative B) Direct
 C) Set associative D) Indirect
20. CUDA is a
 A) Parallel computing platform B) Programming model
 C) Both A & B D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List and briefly define the main structural components of a computer.	2	L*2	1	1
	b) What general categories of functions are specified by computer instructions? Illustrate with example.	4	L2	1	1
	c) With neat sketches, explain various methods for handling multiple interrupts requests raised by multiple devices.	5	L2	1	1
	d) With state diagram briefly define the possible states that define an instruction execution.	5	L3	1	1
2.	a) Explain one address, two address and three address instruction with examples. Using these instructions carry out $C \leftarrow A+B$.	5	L2	1	1
	b) Differentiate straight line sequencing and branching with an example program for each.	6	L2	1	1
	c) Illustrate the working of n-bit ripple carry adder, calculate the gate delay involved in this process.	5	L3	1	1
3.	a) In a carry look ahead addition, explain the generate G_i and propagate P_i functions for stage i. Using this design explain 4 bit carry look ahead adder.	6	L3	2	1
	b) Explain Booth Multiplication algorithm. Apply the booth multiplication algorithm to multiply the signed numbers -5 and 4.	5	L3	2	1
	c) Perform the division of $8 \div 3$ using restoring division.	5	L3	2	1
Unit – II					
4.	a) With a neat diagram explain single bus organization of computer and fundamental concepts.	5	L2	3	1
	b) Draw and explain organization of the control unit to allow conditional branching in the micro program.	5	L3	3	1
	c) Explain the operation of 4-stage pipeline.	6	L2	3	1

5.	a)	Write a note on Data hazards and operand forwarding.	5	L2	4	1
	b)	Write and explain the control sequence for execution of an unconditional branch instruction.	6	L2	4	1
	c)	Compare and contrast the following				
		a. Hard-wired control				
		b. Micro programmed control	5	L2	4	1
6.	a)	Explain cluster computing by considering google data center as an example.	7	L3	4	1
	b)	List the coprocessors used in modern computers and explain the tasks they handle.	3	L2	4	1
	c)	How cache coherence is managed in a NUMA multiprocessor. Illustrate with example.	6	L2	4	1

Unit – III

7.	a)	Demonstrate DMA and its implementation and show how the data is transferred between memory and I/O devices using DMA controller.	6	L2	5	1
	b)	What are the fundamental architectural differences between a CPU and a GPU, and how do these differences influence their respective functions.	7	L2	5	4
	c)	State Amdahl's law.	3	L2	5	1
8.	a)	Explain the role of a CUDA kernel and how it is executed on the GPU. How are threads organized, and what are blocks and grids in this context?	5	L2	5	4
	b)	Discuss the significance of thread indexing using threadIdx and blockIdx, and how these indices are utilized to manage parallelism in CUDA programs.	6	L3	5	4
	c)	What types of memory are available in CUDA, and how do they differ in terms of accessibility, speed, and usage?	5	L2	5	4

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
